SOP - BulkCatalogMigrationAndHygiene - Parent

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Document ID:     SOP - BulkCatalogMigrationAndHygiene - Parent
Naming Format:   [CATEGORY] - [PURPOSE] - [TIER]
Category:        SOP (Standard Operating Procedure)
Purpose:         BulkCatalogMigrationAndHygiene
Tier / Level:    Parent (Master Specification)
Status:          Locked 🔒 (Master Baseline)
Target Audience: TechOps Specialists, Catalog Managers, E-Commerce Merchandisers
Ecosystem Scope: Shopify Admin, Matrixify (Excelify), Google Sheets, 3PL WMS Barcode Scanners
Business Unit:   Synapse Botanicals (Catalog & Data Operations)
```

1. Purpose, Scope & Data Governance

1.1 Purpose

This Standard Operating Procedure (SOP) defines the mandatory protocols for exporting, auditing, mutating, and re-importing product catalog and inventory data in bulk across Synapse Botanicals. Strict adherence to this protocol prevents:

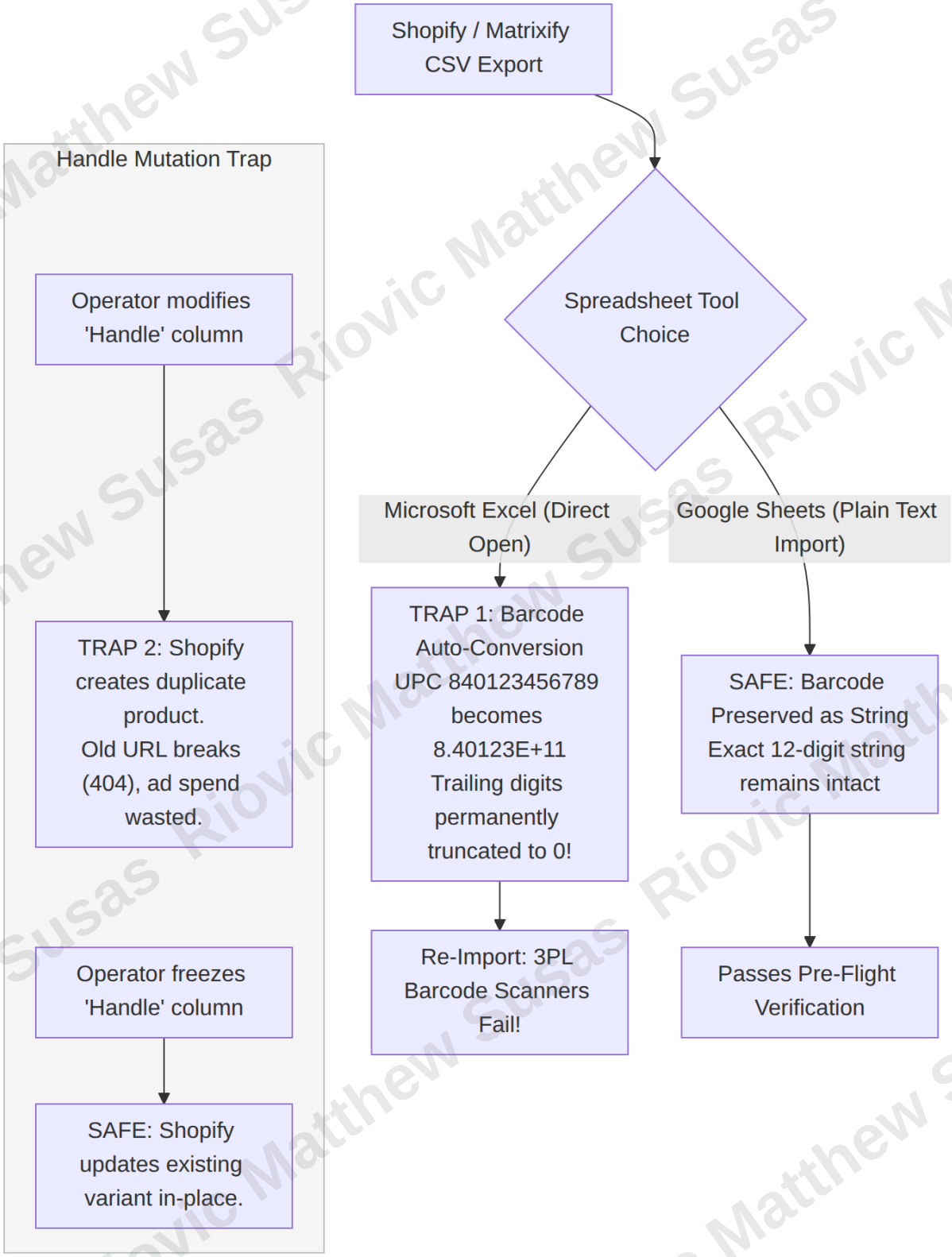
- **Barcode Database Corruption:** Truncation of 12-to-14-digit UPC/EAN barcodes caused by spreadsheet auto-formatting, which immediately disables 3PL warehouse handheld scanners.
- **Orphaned Product URLs & Broken SEO:** Accidental mutation of product `Handle` slugs, breaking active Google search indexation, Meta/TikTok ad tracking links, and customer bookmarks (instant 404 errors).
- **Metafield Type Mismatches:** Corrupted data structures in custom theme sections and headless storefront APIs.
- **Inventory Drift & Ghost Quantities:** Overwriting live Available-to-Sell (ATS) inventory counts during catalog price updates.

1.2 System Scope & Supported Tools

This SOP governs all bulk modifications to product records, variant configurations, cost-of-goods (COGS), and custom metafields using:

- **Matrixify (Excelify):** The authorized enterprise bulk import/export tool for Synapse Botanicals.
- **Google Sheets:** The mandatory spreadsheet editor for preserving raw text strings (Microsoft Excel is prohibited for raw CSV manipulation under Section 2.1).

2. The Two Fatal Traps & Technical Safeguards



2.1 Trap A: The Excel Scientific Notation Barcode Glitch

The Mandatory Safeguard:

1. **Never double-click a CSV to open it in Microsoft Excel.**
2. Use **Google Sheets** (which respects text formatting during import via `File` → `Import` → `Convert text to numbers, dates, and formulas: NO`).
3. If Excel must be used, open a blank workbook → Data → From Text/CSV → Set column data type explicitly to **Text**.

> **CAUTION**
> **THE 12-DIGIT SCIENTIFIC NOTATION CORRUPTION (IEEE 754)**
> When a CSV file is opened directly in Microsoft Excel (double-clicking `.csv`), Excel automatically classifies the 12-to-16-digit `Variant Barcode` column as a numeric data type. Excel cannot natively display integers > 11 digits in standard cells, so it converts `840123456789` into `8.40123E+11`.
> Upon saving, **Excel permanently writes `840123000000` to the file**, destroying the last 6 digits. When re-imported, every product in the catalog fails 3PL barcode scanning, halting warehouse fulfillment nationwide.

2.2 Trap B: The Handle Immutability Rule

The Mandatory Safeguard:
The `Handle` column must remain **read-only** during all routine catalog updates. If a product handle/URL must be changed, an automatic URL forwarding rule must be created immediately so customers clicking the old link are forwarded to the new page instead of seeing a broken link error.

> **WARNING**
> **THE PRODUCT HANDLE IS AN IMMUTABLE PRIMARY KEY**
> In Shopify's database architecture, the `Handle` column (e.g., `synapse-focus-stick-packs-30-pack`) is the **unique system identifier** used to match imported CSV rows to existing products.
>
> - If you modify the handle from `synapse-focus-stick-packs-30-pack` to `synapse-focus-stick-packs-30pk`, Shopify **will not rename the product**.
> - Instead, Shopify creates a **brand new product** with zero reviews, zero sales history, and a new URL. The old product remains orphaned, and all active advertising campaigns targeting the original URL immediately route to a **404 Not Found** page.

> **TIP**
> **What is a URL Redirect (301 Forward)?**
> In Shopify, under **Online Store → Navigation → View URL redirects**, you can set a permanent forwarding rule. If you change a product link from `/products/stick-packs` to `/products/focus-stick-packs`, a redirect ensures that anyone clicking an old Google search result, bookmark, or paid ad is seamlessly redirected to the new link instead of seeing an ugly "404 Page Not Found" error.

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3. SOP - BulkCatalogMigrationAndHygiene - Protocol A: Safe Catalog Export & Data Wrangling

3.1 Matrixify Export Configuration

1. Open Shopify Admin → **Apps** → **Matrixify**.
2. Under the **Export** section, click **New Export**.
3. Select **Format:** `Matrixify: Excel`.
4. In the **Sheets** selection:
   - Check **`Products`**.
   - Under Products options, click the dropdown and select **only the necessary sub-sheets**:
     - `Basic Columns` (Title, Handle, Vendor, Type, Tags, Status)
     - `Variants` (SKU, Price, Compare-At, Cost, Barcode, Weight)
     - `Metafields` (Custom technical attributes)
     - **UNCHECK** `Inventory Levels` (Unless an intentional inventory rebalance is being performed, to avoid overwriting live sales decrements).
5. Click **Export**.

```
Matrixify Export Configuration

Format: Matrixify: Excel (.xlsx)
[✓] Products
    ├── [✓] Basic Columns
    ├── [✓] Variants (Price, COGS, SKU, Barcode)
    ├── [✓] Metafields
    └── [ ] Inventory Levels (Excluded to prevent race conditions)
```

3.2 Spreadsheet Hygiene & Editing Guidelines

1. Open the `.xlsx` export inside **Google Sheets**.
2. Immediately highlight column `Variant Barcode` → Select `Format` → `Number` → **`Plain text`**.
3. Verify that every barcode displays all 12 digits (e.g., `840001040001`).
4. **Permissible Edits:**
   - **`Variant Price`:** Update retail price (e.g., modifying `42.00`).
   - **`Variant Compare At Price`:** Update strikethrough price (e.g., setting to `48.00`).
   - **`Cost per item`** (**`Variant Cost`**)**:** Update unit COGS (e.g., Stick Pack `8.50`, Tincture `12.50`, Starter Bundle `24.50`).

> **NOTE**
> **System Architecture Dependency:**
> Updated `Variant Cost` figures feed directly into the unit economics and financial hazard models documented in **Matrix - OrderStatePointOfNoReturn - Master (Domain 1.2)** – **Section 3.1** and the zone cost models in **Matrix - OrderRoutingCostOptimization - DecisionEngine (Domain 2.2)** – **Section 2.1**.

4. SOP - BulkCatalogMigrationAndHygiene - Protocol B: Custom Metafield Bulk Configuration

Synapse Botanicals relies on custom metafields to render supplement facts panels, active nootropic dosages, and dietary badges on product display pages (PDPs).

4.1 Metafield Naming Syntax in Matrixify

Matrixify requires metafield column headers to follow an explicit syntax:
`Metafield: [namespace].[key] [type]`

Matrixify Column Header Syntax	Shopify Data Type	Production Value Example	Storefront Display Purpose
`Metafield: custom.active_nootropics [string]`	Single-line text	`Lion's Mane, L-Theanine`	Supplement Facts / Active Ingredients
`Metafield: custom.caffeine_mg [number_integer]`	Integer (Whole number)	`100` *(or `0` for Caffeine-Free)*	PDP Caffeine Dose Meter
`Metafield: custom.dietary_badges [list.single_line_text_field]`	List of Strings (JSON Array)	`["Vegan", "Non-GMO", "Gluten-Free"]`	Dietary Certification Badges
`Metafield: custom.serving_count [number_integer]`	Integer (Whole number)	`30`	Servings per Container Display

4.2 Formatting Rules for Lists & JSON Metafields

- When populating `list.single_line_text_field` (e.g., `custom.dietary_badges`), values must be formatted as valid JSON string arrays:
  - **Correct:** `["Gluten-Free", "Vegan", "Sugar-Free"]`
  - **Incorrect:** `Gluten-Free, Vegan, Sugar-Free` *(Will trigger a 422 Unprocessable Entity error during import).*

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5. SOP - BulkCatalogMigrationAndHygiene - Protocol C: Pre-Import Validation & Dry-Run Gate

Before uploading any sheet to production, the operator must complete and sign off on the **5-Point Pre-Flight Verification Gate**:

5-POINT PRE-FLIGHT VERIFICATION GATE		
Verification Checkpoint	Status	Validation Method
1. Barcode String Integrity Check	[ ] PASSED	Regex verification
2. Handle Immutability Audit	[ ] PASSED	VLOOKUP to backup
3. SKU Uniqueness Verification	[ ] PASSED	Conditional Dups
4. Matrixify Command Column Set to MERGE	[ ] PASSED	Header check
5. Full Store Backup Snapshot Exported	[ ] PASSED	Local .xlsx saved

5.1 Verification Checklist Procedures

Check 1: Barcode String Integrity
In a blank column, apply the following formula to test every barcode in row 2:
`=AND(ISNUMBER(VALUE(G2)), LEN(TO_TEXT(G2))>=12, NOT(ISNUMBER(FIND("E+", TO_TEXT(G2)))))`

- If any cell evaluates to `FALSE`, halt immediately. The barcode has been truncated.

Check 2: Handle Immutability Audit
Compare the modified sheet's `Handle` column against the original backup using `=VLOOKUP()`. Confirm zero missing handles (`#N/A`).

Check 3: SKU Uniqueness
Highlight column `Variant SKU` → Apply **Conditional Formatting** → `Format cells if... Custom formula` → `=COUNTIF(E:E, E1)>1`.

- Every SKU must be globally unique across the entire store.

Check 4: Matrixify Command Directive
Ensure Column A is named **`Command`** and set to **`MERGE`**.

- **`MERGE`:** Updates only the fields present in the sheet; leaves unlisted fields intact.
- **`REPLACE` (DANGER):** Wipes and replaces the entire product object, deleting unlisted images and variant metafields.

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6. SOP - BulkCatalogMigrationAndHygiene - Protocol D: Import Execution & Rollback Recovery

6.1 Matrixify Execution Procedure

1. Navigate to **Shopify Admin → Apps → Matrixify**.
2. Under **Import**, click **Upload File** and select your validated `.xlsx` file.
3. Matrixify will parse the file and display an **Import Analysis Summary**:
   - Verify the number of products matches expected count (e.g., `5 Products, 12 Variants`).
   - Verify sheet type displays `Products`.
4. Under **Options**:
   - Set **Import Mode:** `Dry Run (Test without saving)` on initial execution.
   - If Dry Run passes with zero errors, re-run with `Live Import`.
5. Click **Import**.

```
Matrixify Import Analysis

File: Synapse_Catalog_Update_20261017.xlsx
Sheets: Products (5), Variants (12), Metafields (24)

[ Start Dry Run ]        [ Start Import ]
```

6.2 Emergency Rollback Protocol (Disaster Recovery)

If an unintended data mutation occurs (e.g., incorrect pricing or corrupted tags):

1. **Halt Live Import:** Click **Stop Import** inside Matrixify immediately.
2. **Retrieve Pre-Flight Snapshot:** Locate the untouched backup file exported prior to the update (Section 3.1).
3. **Execute Immediate Merge Restore:**
   - Upload the original backup file to Matrixify.
   - Set `Command` column to `MERGE`.
   - Click **Start Import**.
4. **Validate Live Store State:**
   - Open 3 random product pages on the live storefront.
   - Inspect price, compare-at price, and variant dropdowns.
   - Check 3PL portal to confirm inventory sync remains intact.
5. **Log Incident:** File a formal postmortem report detailing the corrupt column header or formula error.

> **NOTE**
> **Audit Trail Reference:**
> For the companion data schema and field mutation risk definitions, consult **Matrix - CatalogDataSchemaAndValidation - Governance (Domain 3.2)**, and for the live verification and dry-run execution logs, consult **Audit - BulkCatalogMutation - RunSheet (Domain 3.3)**.

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